

Chapter 5: Non-stationary processes: models with trend

$$X_t = \text{trend} + \text{stationary component} + \text{noise}$$

(the stationary component is modelling by using the Box-Jenkins methodology: ARMA(p,q) models)

- The trend component has a permanent effect on a time-series: long-run growth rate of real GDP, for example: show graph (enders pag 157)
- The trend can be deterministic or may contain stochastic (random movement: show Pierre example) components (unit root test are helpful at detecting the presence of a stochastic trend).
- We will examine two types of non-stationary processes: TREND-STATIONARY AND DIFFERENCE-STATIONARY PROCESSES

Non-stationary processes: models with trend

- **TREND-STATIONARY PROCESS**: need to regress X_t on time to remove the trend (detrending)

Example:

$$X_t = \beta_0 + \beta_1 t + e_t$$

$$E(X_t) = E(\beta_0 + \beta_1 t + e_t) = \beta_0 + \beta_1 t$$

$$Var(X_t) = Var(\beta_0 + \beta_1 t + e_t) = Var(e_t) = \sigma_e^2$$

This process has a non-stationary mean. The long-term forecast of X_{t+k} will converge to the trend line. By detrending, we are able to remove the deterministic trend.

Shocks to a stationary time series dissipate over time and the series revert to its long-run mean level (shocks are said to be temporary).

Non-stationary processes: models with trend

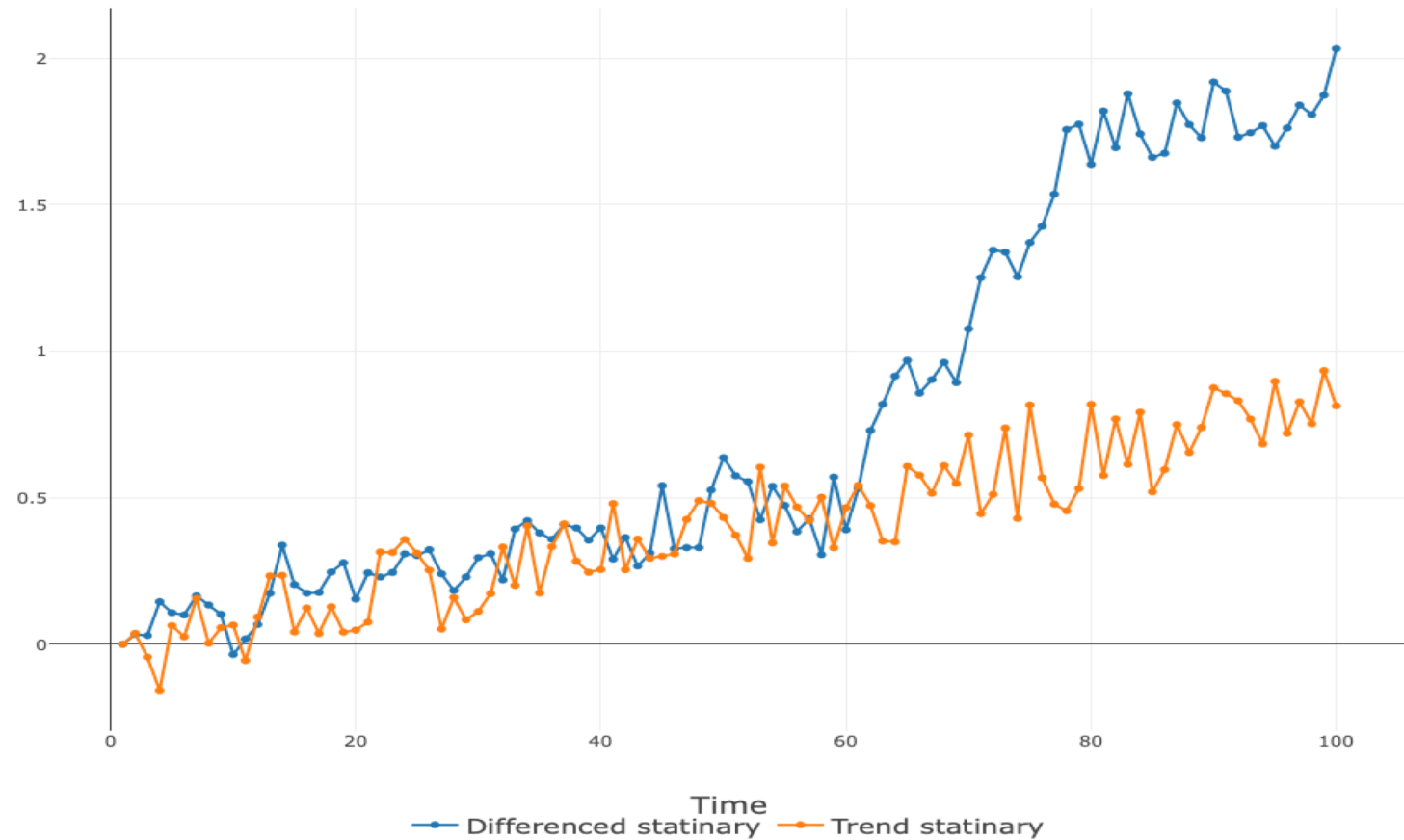
- **DIFFERENCE-STATIONARY PROCESS**: process that exhibit stochastic trend. Whenever there is a shock, the effect of each shock on the intercept is permanent (and randomly): it does not revert to a given mean value.
- Example: The random walk model (an AR(1) with a unit root):

$$X_t = X_{t-1} + e_t$$

- Let's X_0 be the initial condition, this process can be written as:

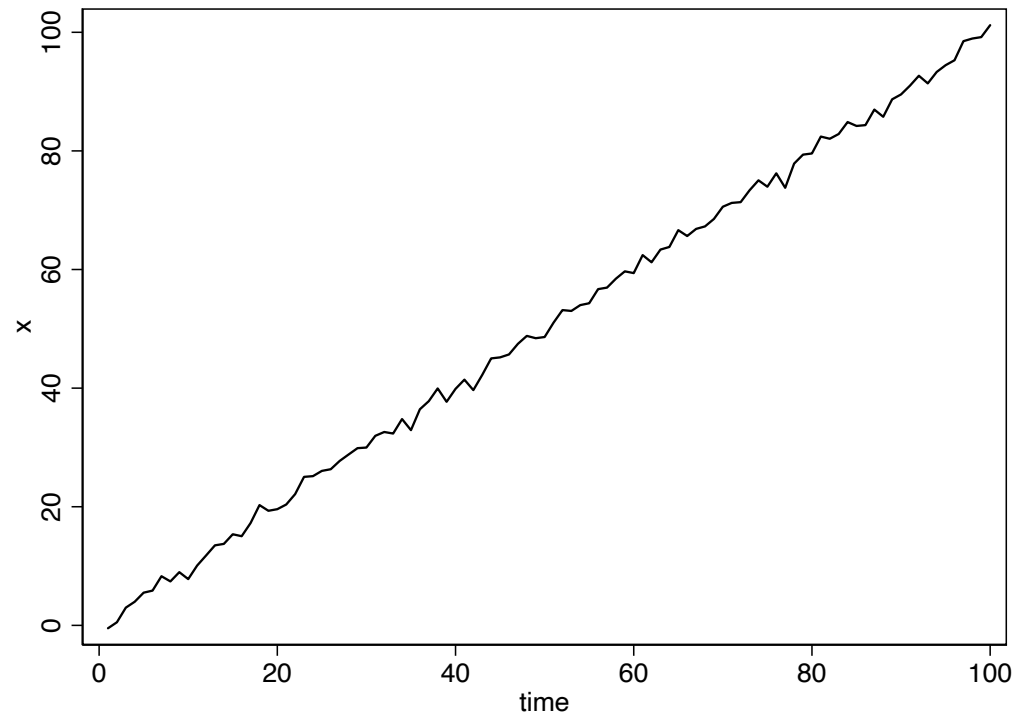
$$X_t = X_0 + \sum_{i=1}^t e_i$$

DIFFERENCED STATIONARY - TREND STATIONARY

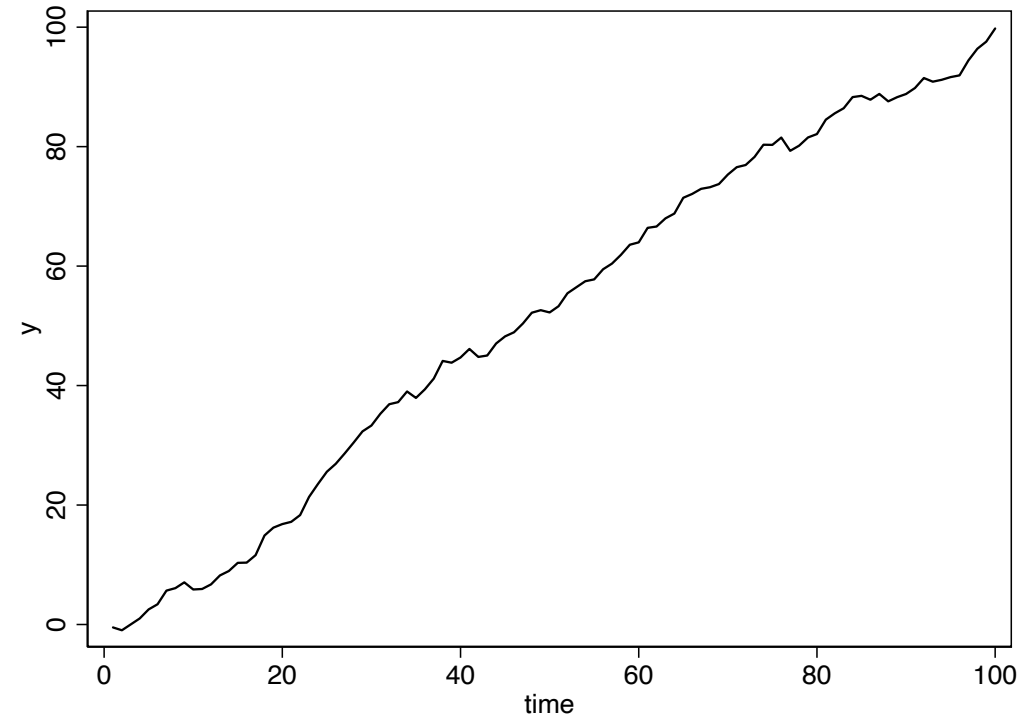


Example: A trend-stationary process (a) and a Difference-stationary process (a random walk with a drift)(b)

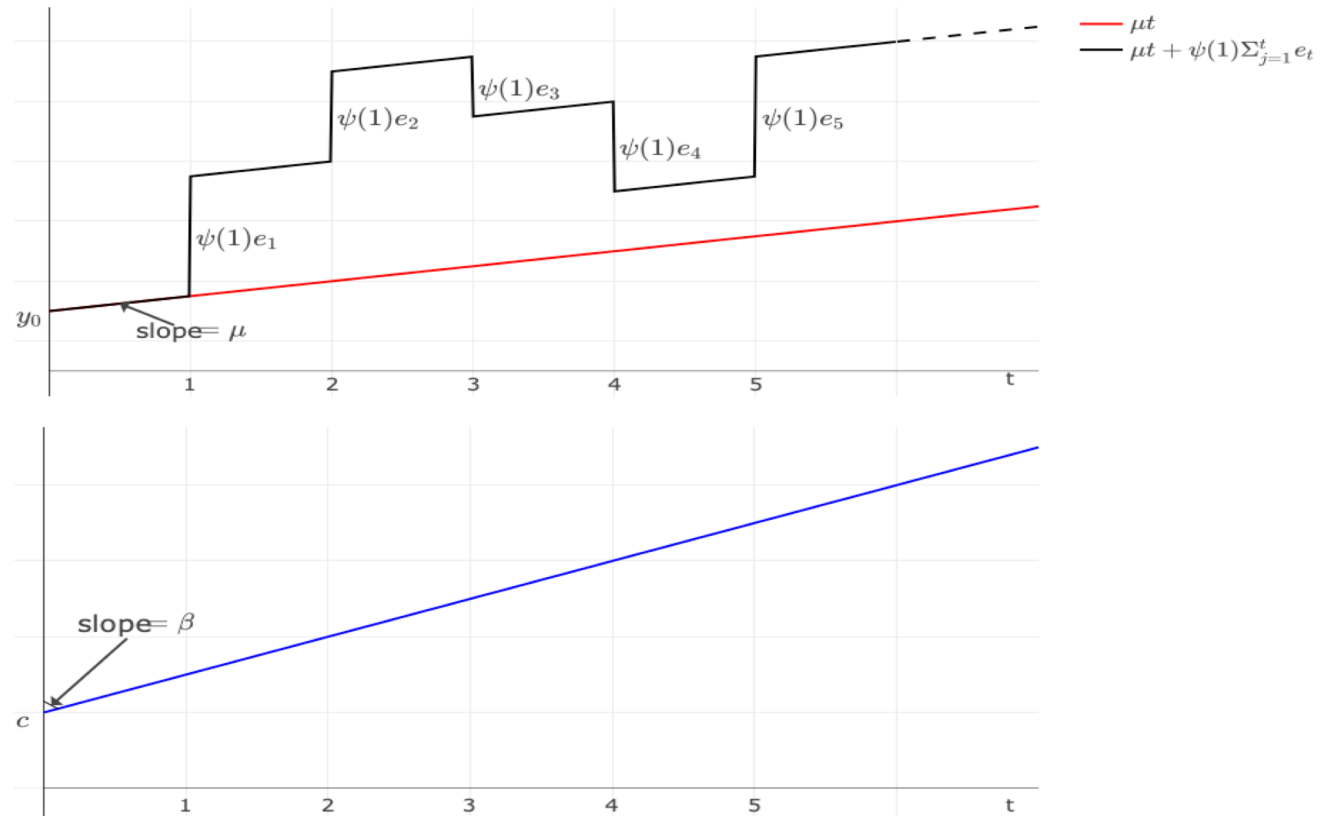
(a)



(b)



Stochastic versus Deterministic Trend



Effect of shocks (e_t) on the level of X_t

- In the first figure above, we can see that the trend function shifts every period, so it is stochastic. Shocks will have a permanent effect on the time series.
- In the second figure above, we have a trend-stationary process. Deviation from the trend function are transitory; shocks (e_t) have no permanent long run effects.

$$X_t = c + \beta t + w_t$$

The series tend to revert to $c + \beta t$

Non-stationary processes: models with trend

- Computing the expected value (conditional to the initial condition):

$$E(X_t|X_0) = X_0 + \sum_{i=1}^t E(e_i) = X_0$$

A random walk process is very unpredictable. The best forecast is the previous period value.

Computing the variance:

$$Var(X_t) = \sum_{i=1}^t var(e_i) = t\sigma^2$$

The variance is a function of time, not constant. This process is not variance-stationary. The autocorrelation also will depend on time.

DIFFERENCE-STATIONARY PROCESS

- An non-stationary process with an stochastic trend is an integrate process. If the process has one unit root, the first difference will be stationary.

$$\Delta X_t = e_t$$

Where ΔX_t is a stationary process.

Another example: A Random walk with drift (β_0 , a constant term is added) process:

$$X_t = \beta_0 + X_{t-1} + e_t$$

- Let's $X_0 = 0$, this process can be written as:

$$X_t = \beta_0 t + \sum_{i=1}^t e_i$$

A Random walk with drift

$$E(X_t) = \beta_0 t + \sum_{i=1}^t E(e_i) = \beta_0 t$$

A random walk process is very unpredictable. The best forecast is the previous period value.

Computing the variance:

$$Var(X_t) = var(\beta_0 t) + \sum_{i=1}^t var(e_i) = t\sigma^2$$

Taking the first difference:

$$\Delta X_t = \beta_0 + e_t$$

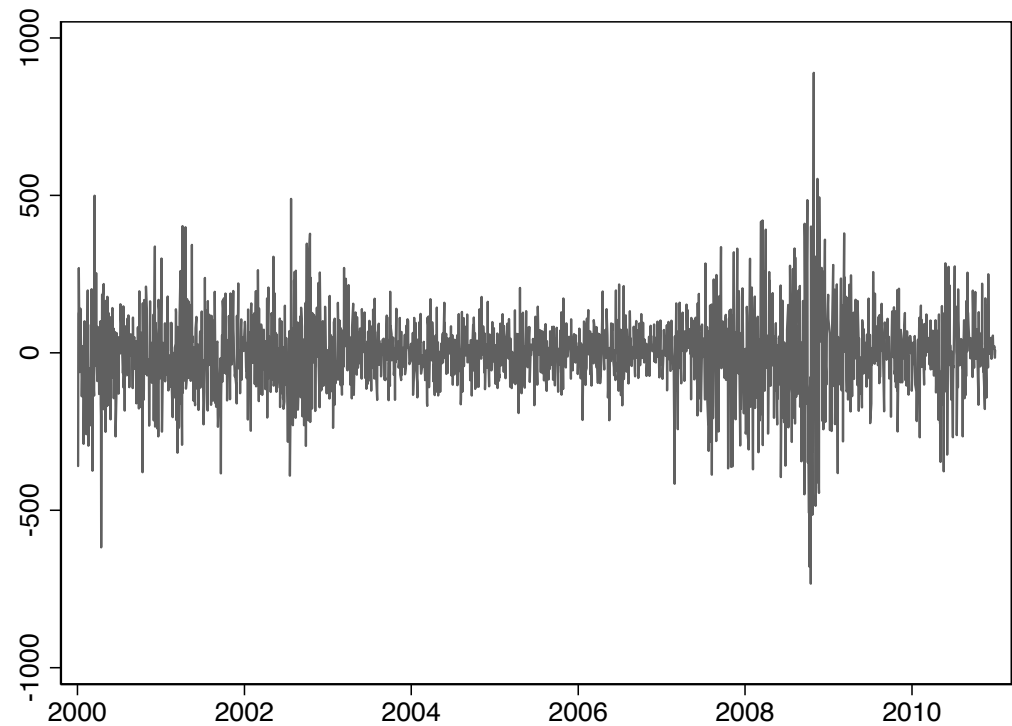
ΔX_t is a stationary process: $E(\Delta X_t) = \beta_0$, $Var(\Delta X_t) = \sigma^2$, and $Cov(\Delta X_t) = 0$

Example: (a) Down Jones Industrial avg.(in level) b) The first difference of the Down Jones Industrial avg.

(a)



(b)



Mistakenly differencing a Trend Stationary Process

- First differencing a trend stationary process induces an MA unit root.
- Example:

$$X_t = \beta_0 + \beta_1 t + e_t$$

$$\Delta X_t = (\beta_0 + \beta_1 t + e_t) - (\beta_0 + \beta_1(t-1) + e_{t-1})$$

$$\Delta X_t = \beta_1 + e_t - e_{t-1}$$

*Canadian Inflation paper

Mistakenly detrending a unit root process

- Durlauf and Phillips (1988) show that mistakenly detrending a unit root process invalidates most hypothesis testing and induces spurious cyclicalities or seasonality in the data.
- Overdifferencing a process will not only induce a unit root in the MA process, it will also double the variance of the process and introduce negative autocorrelation into the data.
- Show examples on pages 110/111 and 112
- Replicating Granger and Newbold (1974) paper.